

Cutting Plane Methods

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1 GLS Oracle Model

1.1 Definitions

Definition 1 (Oracle Model). *The following are algorithmic problems for closed convex set $K \subseteq \mathbb{R}^n$:*

MEMbership($x \in \mathbb{R}^n, K \subseteq \mathbb{R}^n$):

Output: YES if $x \in K$; NO otherwise.

*SEP*aration($x \in \mathbb{R}^n, K \subseteq \mathbb{R}^n$):

Output: YES if $x \in K$;

otherwise output separating hyperplane w such that $\langle w, x \rangle > \sup_{y \in K} \langle w, y \rangle$;

*OPT*imization($w \in \mathbb{R}^n, K \subseteq \mathbb{R}^n$)

Output: $h_K(w) = \sup_{x \in K} \langle w, x \rangle$

and optimizer $x_ \in K$ such that $h_K(w) = \langle w, x_* \rangle$;*

*VAL*idity($w \in \mathbb{R}^n, b \in \mathbb{R}, K \subseteq \mathbb{R}^n$):

Output: YES if $\langle w, x \rangle \leq b$ for all $x \in K$; NO otherwise.

*VIOL*ation($w \in \mathbb{R}^n, b \in \mathbb{R}, K \subseteq \mathbb{R}^n$):

Output: YES if $\langle w, x \rangle \leq b$ for all $x \in K$;

otherwise output $x \in K$ such that $\langle w, x \rangle > b$.

We can also consider *approximate* versions of all these oracles. For the formal definitions, see [GLS 2.1]. Many of the algorithms we study in this class can be thought of as reductions between these algorithmic problems.

1.2 Relations between Oracles

The various oracles for K and its polar dual K° are related.

Proposition 2. *In the following, we assume $B(0, \varepsilon) \subseteq \text{int}(K)$ and $B(0, \varepsilon) \subseteq \text{int}(K^\circ)$.*

- *OPTimization oracle for K can be (approximately) implemented using VIOLation or VALidity oracle for K ;*
- *MEMbership oracle for K is equivalent to VALidity oracle for K° ; and vice-versa $MEM(K^\circ)$ is equivalent to VAL(K);*

- *SEParation oracle for K° can be implemented using a VIOLation oracle for K ; $VIOL(K)$ can be implemented using $SEP(K^\circ)$ and one of $MEM(K)$ or $VAL(K^\circ)$.*

Proof:

- In the following we use VAL if we only need the optimum value, and $VIOL$ if the optimizer is also required. Using the $VAL/VIOL$ oracles we can check if $\exists x \in K : \langle w, x \rangle \geq b$. Therefore, to compute $h_K(w) = \sup_{x \in K} \langle w, x \rangle$ approximately, we perform binary search over b .
- We first implement $MEM(K^\circ)$ using $VAL(K)$. Given input w we want to check if $w \in K^\circ$ iff $\sup_{x \in K} \langle w, x \rangle \leq 1$. For this we return the output of $VAL(w, 1, K)$.

Conversely, we want to implement $VAL(K)$ using $MEM(K^\circ)$. So given an input (w, b) we want to check if $\forall x \in K : \langle w, x \rangle \leq b$. First we use that $B(0, \varepsilon) \subseteq K$ so if $b \leq 0$ we output NO. Otherwise note that $v \in K^\circ$ iff $\sup_{x \in K} \langle v, x \rangle \leq 1$. So we can return the output of $MEM(w/b, K^\circ)$.

The dual statement, swapping K and K° , follows due to bi-duality $(K^\circ)^\circ = K$.

- We first implement $SEP(K^\circ)$ using $VIOL(K)$. Given input w we want to check if $w \in K^\circ$, and if not to output a separating hyperplane. Note $w \in K^\circ$ iff $\sup_{x \in K} \langle w, x \rangle \leq 1$, so we query $VIOL(w, 1, K)$; if the output is YES, we return YES; if the output is NO then the oracle returns $x \in K$ such that $\langle w, x \rangle > 1 \geq \sup_{v \in K^\circ} \langle v, x \rangle$, where the last step is by definition of K° , so we can output x as our separating hyperplane for K° .

Conversely, we want to implement $VIOL(K)$ using $SEP(K^\circ)$. So given an input (w, b) we want to check if $\sup_{x \in K} \langle w, x \rangle \leq b$, and if not to output $x \in K$ such that $\langle w, x \rangle > b$. First we use that $B(0, \varepsilon) \subseteq K$ so if $b \leq 0$ we output $0 \in K$ as $\langle w, 0 \rangle = 0 > b$. Otherwise we query $SEP(w/b, K^\circ)$; if the output is YES then $\sup_{x \in K} \langle w/b, x \rangle \leq 1$ so we return YES; if the output is NO then the oracle returns separating hyperplane $x \in \mathbb{R}^n$ such that $\langle w/b, x \rangle > \sup_{v \in K^\circ} \langle v, x \rangle$. If we knew this right hand side $h_{K^\circ}(x) = \sup_{v \in K^\circ} \langle v, x \rangle$ then we could output $y := x/h_{K^\circ}(x) \in K^{\circ\circ} = K$ with $\langle w, y \rangle > b$ as our violating point. But we can approximately compute this value using binary search with the VAL idity oracle for K° as shown in the first part. Equivalently, since $0 \in \text{int}(K)$, there is a point $y \in [0, x] \cap \partial K$ at the intersection of the line $[0, x]$ with the boundary of K , and this also gives a violating point $\langle w, y \rangle > b$ ($y \in \partial K$ implies $\sup_{v \in K^\circ} \langle v, y \rangle = 1$). And this boundary point we can approximately compute using binary search with the MEM bership oracle for K .

□

1.3 Function Oracles

What is the relation to function oracles? In particular, recall that we prove duality of convex functions by reducing to duality of the epigraph, which is a convex set.

Definition 3 (Fenchel dual). *For convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the Fenchel dual is*

$$f^*(w) := \sup_y \langle w, y \rangle - f(y).$$

Lemma 4. *For differentiable closed convex f , the supporting hyperplane for the epigraph of f at $(x, f(x))$ is given by slope $(\nabla f(x), -1)$ and value $f^*(\nabla f(x)) = \sup_y \langle \nabla f(x), y \rangle - f(y)$.*

Similarly, the supporting hyperplane for the sub-level set $L_{f(x)} := \{y \in \mathbb{R}^n \mid f(y) \leq f(x)\}$ is given by slope $\nabla f(x)$ and value $\langle \nabla f(x), x \rangle$.

Both these statements can be generalized to non-differentiable f using subgradients.

Proposition 5. *In the following, we assume $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a closed convex function.*

- *EVALuation oracle for f can be implemented using MEMbership oracle for $\text{epi}(f)$;*
- *GRADient oracle for f can be implemented using SEParation oracle for $\text{epi}(f)$;*
- *OPTimization oracle for $\text{epi}(f)$ is equivalent to EVALuation for the Fenchel dual $f^*(w) := \sup_y \langle w, y \rangle - f(y)$.*

2 Cutting Plane Methods

Our eventual goal is to find an algorithm to solve convex programs $\min_{x \in K} f(x)$. Cutting plane methods solve this using just SEParation oracle for K and EVAL and GRAD oracles for f . We first show how to solve the much simpler FEASibility problem: given SEParation oracle for K , either find $x \in K$ or output that K is ‘small’. Note that linear optimization over K can be reduced to this: test FEASibility for

$$K' := K \cap \{x \mid \langle w, x \rangle \geq b\}.$$

and run binary search on b . Note that SEP for K' can be easily implemented using SEP for K and testing the linear inequality $\langle w, x \rangle \geq b$.

The intuition for cutting plane methods comes from binary search: say we are attempting to find some point $x \in [0, 1]^n$, and we have oracle access to coordinate queries $y_i \geq x_i$ or $y_i \leq x_i$. Then in each iteration, we should query the center of the remaining grid and eliminate the elements with chosen coordinate too large or small, depending on the output of the oracle. This procedure is optimal in the worst case, as it eliminates the maximum possible options in each iteration. This is the correct intuition for the Center of Gravity method, described below, which generalizes the procedure to arbitrary convex $K \subseteq \mathbb{R}^n$. In general, it is not clear how to query the ‘center’ of a convex body, so the next algorithm will maintain an Ellipsoid in each iteration to contain the feasible set, and update according to the separation oracle.

3 Center-of-Gravity Method

We first describe the algorithm to solve feasibility. In each iteration, we maintain a feasible convex set $K_t \supseteq K$; we query the separation oracle with the Center of Mass $c_t := \mathbb{E}_{x \sim K_t}[x] = \int_{x \in K_t} x dx$; if $c_t \in K$ then we are done; otherwise we get $\langle w, c_t \rangle > \sup_{x \in K} \langle w, x \rangle$; therefore, we know K is contained in the following halfspace

$$H_t := \{x \mid \langle w, x \rangle \leq \langle w, c_t \rangle\},$$

so we update accordingly $K_{t+1} := K_t \cap H_t$. We claim that this solves feasibility:

Theorem 6. *Given $K \subseteq B(0, R) \subseteq \mathbb{R}^n$ via separation oracle, the center-of-gravity method requires $O(n \log(R/\varepsilon))$ iterations to either*

1. *find $x \in K$;*
2. *certify K does not contain a ball of radius ε , or certify $\text{vol}(K) \leq \varepsilon^n$.*

The key step in the analysis is the following beautiful result from convex geometry:

Theorem 7 (Grunbaum's Theorem). *For convex $K \subseteq \mathbb{R}^n$, let $c := \mathbb{E}_{x \in K} x$ be the center-of-gravity of K . Then for any hyperplane $H \ni c$, the two halfspaces H_+, H_- satisfy*

$$\max\{\text{vol}(K \cap H_+), \text{vol}(K \cap H_-)\} \leq \text{vol}(K)(1 - 1/e).$$

We do not prove this in this class but refer to the excellent survey of Keith Ball: An Elementary Introduction to Modern Convex Geometry. With this result, the analysis is straightforward.

Proof: [Proof of theorem 6] If in any iteration we find $c_t \in K$ then we are done. Otherwise, in each iteration we reduce the volume by the factor stated above. Therefore in $T = O(n \log(R/\varepsilon))$ iterations we have

$$\text{vol}(K_T) \leq \text{vol}(K_0)(1 - 1/e)^T \leq \text{vol}(B(0, R)) \exp(-n \log(R/\varepsilon)) \leq \text{vol}(B(0, \varepsilon)),$$

where we used that $K \subseteq B(0, R)$. □

In the following section, we show that this is in fact the optimal query complexity possible for an algorithm using just a SEParation oracle. Of course, as stated it is not at all clear how to compute the center-of-gravity of K_t , and it turns out this is at least as hard as optimizing over K_t . Therefore in the following sections we will study the Ellipsoid algorithm, which requires more oracle queries but can be efficiently updated.

We next describe a very similar algorithm to solve general convex programs. In each iteration, we still maintain a feasible convex set $K_t \supseteq K$ and query the center c_t ; if $c_t \notin K$ then we update $K_{t+1} = K_t \cap H_t$ just as in the previous algorithm; if $c_t \in K$ then we query the EVALuation and (sub)-GRADient oracle for f , to compute $(f(c_t), g)$; Finally we update $K_{t+1} = K_t \cap H_g$ where

$$H_g := \{x \mid \langle g, x - c_t \rangle \leq 0\}$$

i.e. the halfspace in the negative (sub-)gradient direction.

Theorem 8. *Given $K \subseteq B(0, R) \subseteq \mathbb{R}^n$ via SEParation oracle, and convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$ via EVALuation and GRADient oracle, such that $\max_{x \in K} f(x) - \min_{x \in K} f(x) \leq F$; The center-of-gravity method requires $O(n \log(RF/\varepsilon))$ iterations to either*

1. *find $x \in K$ such that $f(x) \leq \min_{y \in K} f(y) + \varepsilon$;*
2. *certify $\text{vol}(K) \leq \varepsilon^n$.*

The proof rests on the following claim, showing the negative gradient gives a good update.

Claim 9. *For closed convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $x \in \text{dom}(f)$, any subgradient $g \in \partial f(x)$ gives a supporting hyperplane for the sub-level set*

$$L_{f(x)} := \{y \mid f(y) \leq f(x)\}.$$

We leave the proof as an exercise. But intuitively the negative subgradient halfspace always contains the optimizer $x_* := \arg \min_{x \in K} f(x)$, so our update maintains a body containing x_* . With this in hand, the analysis follows:

Proof: [Proof of theorem 8] We still manage to decrease the volume by a constant factor in each iteration, due to Grunbaum's theorem. And the claim above shows that $K_t \ni x_*$ for all t . If in every iteration $c_t \notin K$, then by the previous feasibility analysis we can certify small volume in $T = O(n \log(R/\varepsilon))$ iterations. Otherwise, we show that when the volume gets small enough, one of the queried center points must have small function value.

For this we analyze the follows approximate optimal set:

$$X_\delta := (1 - \delta)x_* + \delta K = \{(1 - \delta)x_* + \delta x \mid x \in K\}.$$

Note that we can bound the function on this set as, for $x \in K$,

$$f((1 - \delta)x_* + \delta x) \leq (1 - \delta)f(x_*) + \delta f(x) \leq (1 - \delta)f(x_*) + \delta(f(x_*) + F) \leq f(x_*) + \delta F,$$

where the first step was by convexity of f , and in the second step we used our bound $\max_{x \in K} f(x) - \min_x f(x) \leq F$. Note that $\text{vol}(X_\delta) = \text{vol}(\delta K) = \delta^n \text{vol}(K)$. Therefore if we reach T such that $\text{vol}(K_T) < \text{vol}(X_\delta)$ then we must have some point $x \in X_\delta$ that is not contained in our current feasible set $x \notin K_T$. Therefore in some iteration we must have queried $c_t \in K_t$ and cut off some part of X_δ . $x_* \in K_T$ always by the above claim, so we can choose $x_\delta \in X_\delta - K_t$ to be a cut-off point on the line from $x_* \rightarrow c_t$, and we can write $c_t = (1 - \lambda)x_* + \lambda x_\delta$ for some $\lambda \in [0, 1]$. Then

$$f(c_t) \leq (1 - \lambda)f(x_*) + \lambda f(x_\delta) \leq f(x_\delta) \leq f(x_*) + \delta F,$$

again by convexity and the derived bound on $f(x_\delta \in X_\delta)$. Choosing $\delta \lesssim \varepsilon/F$ gives the result. $\square$

We will show in the following that this achieves the optimal *query* complexity with respect to the separation oracle. Of course we have not specified how to actually find the center of gravity of a given polytope. It turns out that this task is very computationally expensive (about as difficult as solving the optimization problem itself). Another variant that is more efficient uses the maximum inscribed ellipsoid of the current feasible region and queries its center. Computing the maximum inscribed ellipsoid is also difficult, but we maintain optimal query complexity with this algorithm. The analysis shows this improves the volume of the *max inscribed ellipsoid* by a constant factor in each iteration (not necessarily the volume of the feasible set). In the remainder we will discuss the Ellipsoid method: this does not keep track of all previous separating halfspaces, but instead maintains and updates a single ellipsoid that contains the feasible region. This method gives up on optimal query complexity, but each iteration can be implemented efficiently.

4 Lower Bound

Theorem 10. *Given $K \subseteq B(0, R) \subseteq \mathbb{R}^n$ via SEPARATION oracle, any algorithm solving the FEASIBILITY problem requires $\Omega(n \log(R/\varepsilon))$ oracle calls to the SEP oracle.*

Proof: [Proof Sketch] Consider the grid example $K = x + [-\varepsilon, \varepsilon]^n \subseteq [0, 1]^n$, and note that for each query, the adversary can always choose a separating hyperplane (even a coordinate hyperplane) to keep $\Omega(1)$ fraction of the mass. $\square$